HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Administered and optimized enterprise engineering tools including **PDM/PLM systems**, ensuring template alignment with global design standards for an engineering library of over 50,000 components.
- Developed and deployed **Java Spring Boot REST APIs** for cross-service data exchange, incorporating request validation and circuit-breaker patterns to manage ~1,500 requests per minute.
- Authored **Python scripts** to automate data extraction and transformation for scheduled reporting, reducing manual effort for the operations team by approximately 15 hours per week.
- Collaborated with multi-disciplinary teams (Engineering, Procurement, Manufacturing) to gather product requirements and address design pipeline challenges, embedding standard templates from concept to installation.
- Engineered fault-tolerant **Java consumers** for a financial-services operations dashboard, parsing inbound transaction messages and surfacing alerts for approximately 3,200 failed payments per week across monitored queues.
- Containerized applications using **Docker** and deployed them to an AWS EKS cluster, managing configurations and resource limits for predictable rollouts.
- Assisted Data Analysts with data collection for the development of **Power BI dashboards** tracking team performance and engineering metrics, ensuring data accuracy for approximately 10 key performance indicators.

SKILLS

Engineering Platforms – PDM, PLM, CAD, BIM, SolidWorks, Autodesk Inventor, Revit, NavisWorks, Onshape

Programming Languages – Java, Python, SQL, R, TypeScript

Frameworks & Libraries – Spring Boot, Hibernate, JPA, pandas, NumPy, scikit-learn, Flask, FastAPI, NestJS

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, Linux

Data & Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, SQL (PostgreSQL, MySQL, Oracle)

Methodologies – Agile, Scrum, CI/CD, Change Management, Design for Manufacturing and Assembly (DFMA)

PROJECTS

Engineering Design Automation Tools (HCL Technologies)

- Developed **Python-based automation scripts** to generate CAD drawings and Bill of Materials (BOM) templates, reducing manual creation time for standard product configurations by 40%.
- Implemented a **RESTful API using Flask** to integrate the automation tool with the PDM system, enabling automated data synchronization for approximately 500 product variations.
- Authored technical documentation and user guides for the automation tools, facilitating adoption by the engineering teams and supporting internal training sessions.

Internal Systems Audit Support (HCL Technologies)

- Assisted in organizing and executing internal engineering systems audits, focusing on adherence to approved workflows and quality standards across 3 key engineering platforms.
- Collected and validated data from **PDM/PLM systems** and engineering logs to support the development of **Power BI dashboards** for tracking team performance and engineering metrics.
- Performed ad-hoc **SQL queries** against enterprise databases to extract and reconcile data for audit reporting, ensuring data integrity for over 10,000 engineering records.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)